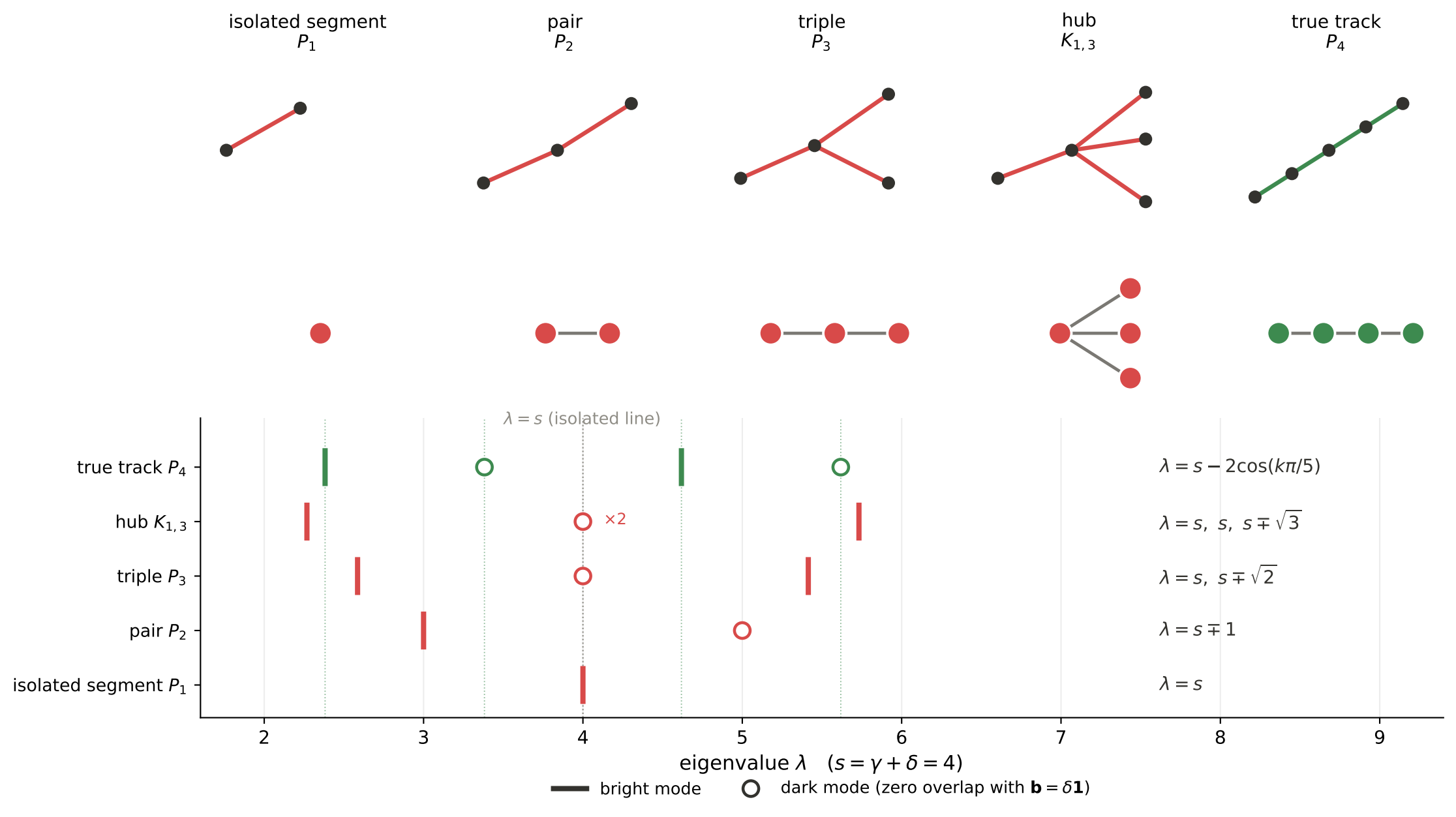
isolated segment
P_1
pair
P_2
triple
P_3
hub
$K_{1,3}$
true track
P_4
$\lambda = s$ (isolated line)
true track P_4
hub $K_{1,3}$
triple P_3
pair P_2
isolated segment P_1
×2
$\lambda = s - 2\cos(k\pi/5)$
$\lambda = s,\ s,\ s \mp \sqrt{3}$
$\lambda = s,\ s \mp \sqrt{2}$
$\lambda = s \mp 1$
$\lambda = s$
2
3
4
5
6
7
8
9
eigenvalue λ $(s = \gamma + \delta = 4)$
bright mode
dark mode (zero overlap with $\mathbf{b} = \delta\mathbf{1}$)